Demand side data

We will assume we have a baseline 50 MW load. (IT Load) and PUE values with this function:

The PUE (Power Usage Effectiveness) Values

Outside Temp (Tamb)	Cooling Mode	PUE Value	ISO Impact
< 10°C (50°F)	Free Cooling (Economizer)	**1.12**	Common in PJM/MISO Winters
10°C to 25°C	Standard Liquid Cooling	**1.20**	Typical Spring/Fall logic
25°C to 35°C	Hybrid / Assisted Cooling	**1.32**	Summer in PJM; Days in ERCOT
> 35°C (95°F)	Max Mechanical Chilling	**1.45**	Summer Afternoons in ERCOT

Supply side data

- **Solar Photovoltaic (PV) Generation Modeling**

To accurately model the Behind-the-Meter (BTM) supply, we do not rely on static capacity factors. Instead, we use a dynamic physics-based model that calculates hourly power output based on Global Horizontal Irradiance (GHI) and Ambient Temperature sourced from the NREL National Solar Radiation Database (NSRDB). We will use the API:

https://nsrdb.nrel.gov/data-viewer

Commented [AΦ1]: Do they have forecast? Or use 2024

The hourly power output will then be calculated using the following governing equation:

$$P_{out} = P_{DC} \times \left(\frac{POA}{1000}\right) \times [1 + \gamma \times (T_{cell} - 25)] \times \eta_{sys}$$

Where:

- P_{DC}*: The installed capacity in MW (the decision variable in our optimizer).*
- POA*: Plane of Array irradiance*
- γ*: Temperature coefficient of the PV modules (assumed at* ***-0.35%/°C*** *for 2026 high-efficiency N-type cells).*

- T_{cell}: *Estimated cell temperature, calculated as Tambient + (POA * 0.025).*
- η_{sys}: *System derate factor (assumed at **0.85**) to account for inverter losses, soiling, and wiring.*

We will select indicative/representative for each ISO.

- **Battery Energy Storage System (BESS) Modelling:**

B (Battery Energy Capacity in MWh): The total energy storage capacity. (We can use ambient temperature if we want to include PHVAC losses which gives a per ISO difference)

For every hour, we should calculate the new energy level in the battery:

$$E_t = E_{t-1} + \left[P_{charge,t} \times \left(\eta_{ch} \times \eta_{temp,t}\right)\right] - \left[\frac{P_{discharge,t}}{\eta_{temp}\eta_{dis}}\right]$$

$$0.20B \leq E_t \leq B(Energy\ limits/Depth\ of\ Discharge)$$

- **Gas modelling:**

The gas supply is a gap – filler.

$$P_{gas,t} = min(G, unmet\ demand\ at\ t)$$

- **G:** The **Decision Variable** for installed gas capacity (MW).
- **Unmet Demand:** The deficit remaining after Solar_t and Battery Discharge are applied to the load.

$$Fuel\ consumption = Pgas_t * Heat\ Rate$$

- **Grid imports modelling:**

For times where demand cannot be satified from the microgrid we will import from the grid:

$$Pgrid$$

How will our hourly simulation look like:

Hour (t)	Temp (∘C)	PUE	Total Load (Lt)	Solar Gen (Ps)	Grid Import (Pgr)	Battery SOC (Et)	Gas Gen (Pg)
001	12	1.15	57.5 MW	0.0 MW	10.0 MW	420 MWh	47.5 MW
002	38	1.40	70.0 MW	95.0 MW	0.0 MW	445 MWh	0.0 MW
...	...	...	...	...	...	...	...
8760	5	1.12	56.0 MW	0.0 MW	10.0 MW	110 MWh	46.0 MW

Decision Variable	Unit
S (Solar) installed capacity	MW
B (Battery)	MWh
G (Gas)	MW
Pgrid	MW

Economics:

For the LCOEs of each technology we will use:

- Lazard 18.0 (2025) report which provide a national average: https://www.lazard.com/media/5tlbhyla/lazards-lcoeplus-june-2025-_vf.pdf
- NREL ATB 2025 capital cost multipliers https://atb.nrel.gov/
- **EIA 2026** regional fuel price forecasts https://www.eia.gov/outlooks/steo/
- EIA AEO 2025 (for capacity factors)

Methodology (probably):

Regional LCOE figures were derived by localizing the Lazard 18.0 national baseline with NREL ATB 2025 capital cost multipliers and EIA 2026 regional fuel price forecasts. This accounts for the significant variance in solar irradiance and gas delivery costs between ERCOT (high yield/low fuel) and PJM (low yield/high fuel).

And we end up with the equation: (simplified for our ease)

$$sLCOE = \frac{(E_{sol} * LCOE_{sol}) + (E_{gas} * LCOE_{gas}) + (E_{bat} * LCOS_{bat}) + (E_{grid} * LCOE_{grid})}{Total\ Energy\ Consumed\ Annual}$$

- Ex Total MWh provided by that specific source (Solar, Gas, Grid) over 8,760 hours.

- Ebat: Total MWh discharged by the battery.
- LCOEs The standalone values we already have from lazards report

The grid LCOE formula

$$Annual_{Grid_{Cost}} = \sum_{t=1}^{8760} (Pgrid_t * LMP_t) + \sum_{m=1}^{12} (Capacity_{Fee_m} + Demand_{Charge_m}) + Fixed_TD$$

$$LCOE_{grid} = \frac{Annual_{Grid_{Cost}}}{\sum_{t=1}^{8760} Prid_t}$$

We need to get:

- the hourly LMPs for each ISO
- Capacity_Fees
- Demand Charges
- Fixed T&D costs for our collocated facility

So first objective is to meet demand.

Second objective so to have a low sLCOE.

We can start with E_grid = 0 but if we want to make it more realistic we must add possible grid imports.

We will do the simulation for 2024:

PJM: https://dataminer2.pjm.com/feed/rt_hrl_lmps

ERCOT: https://www.ercot.com/mktinfo/rtm

MISO: https://www.misoenergy.org/markets-and-operations/real-time--market-data/market-reports/#t=10&p=0&s=MarketReportPublished&sd=desc

CAISO: https://oasis-bulk.caiso.com/

SPP: https://portal.spp.org/pages/hourly-spp

NYISO: https://www.nyiso.com/custom-reports

ISO-NE: https://www.iso-ne.com/isoexpress/web/reports/pricing/-/tree/lmps-rt-hourly-final